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A prediction model for the oceanauts' cognitive performance based on the mental workload of typical tasks

Mengya Zhu^{1*}, Jingluan Wang^{2,3}, Xian Zhang^{2,3} and Dengkai Chen^{2,3}

Abstract

High-level cognitive abilities are essential for the oceanauts to successfully complete the tasks, such as assessing cruising positions, operating interface commands, and responding to voice instructions. However, it is still unknown which variables impact oceanauts' cognitive performance and how, and how best to create and implement analytic approaches to predict oceanaut's personalized cognitive performance. As a result, we attempted to conduct cognitive tests trials on four typical activities among oceanauts. Oceanaut cognition characteristics were collected in order to create a quantitative model of cognitive performance. We utilised Radial Basis Functions to predict the oceanauts' cognitive performance and investigated the impact of mental workload, personal characteristics, and comfort on their cognitive performance. The findings indicated that establishing an appropriate level of mental workload for the visual, auditory, cognitive, and psychomotor channels can enhance the oceanaut's cognitive performance of individuals engaged in ocean exploration. It is worth noting that the optimal values for mental workload vary across these channels. Furthermore, the presence of suitable comfort, appropriate age, and a relatively higher level of educational background contribute to the enhancement of cognitive performance among oceanauts.

Keywords Cognition performance, Mental workload, Prediction model, The oceanauts, Typical tasks

Introduction

The manned submersible is an essential apparatus utilised for scientific research and exploration in the depths of the ocean [1]. The oceanaut, in their capacity as the pilot of a manned submersible, assumes a prominent role in executing deep-sea diving missions [2, 3]. However, the operating environment of the manned submersible is harsh, and the tasks are heavy [4]. The internal diameter

of manned submersibles is typically only 2 m, requiring accommodation for three crew members along with essential operational equipment [5]. Consequently, the available working space is extremely constrained. Moreover, the limited battery power supply places significant restrictions on the use of environmental control systems. As a result, oceanauts often face challenging interior conditions, such as elevated temperatures, high humidity, restricted movement space, and excessive noise and vibration for a long time. As a notable achievement for mankind to challenge the limits of ocean survival and operations [6], the operating duration of a manned submersible can extend to 12 h or longer [7], with oceanauts typically needing to stay focused on diving tasks in darkness for 8 to 10 h [2, 8].

Research findings indicate that the reduced performance of employees can lead to significant safety risks

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[9]. 70–80% of manned deep submergence accidents are related to human factors according to the statistics of relevant sea trials [10, 11]. One of the leading causes of errors among oceanauts is that the performance of the oceanauts declines over time [12]. The cognitive performance of oceanauts plays a crucial role in ensuring the successful completion of operational tasks. Considering scenarios involving single-person operations and the need for individualized adjustment strategies, developing a personalized predictive model for the cognitive state of oceanauts has become a critical concern.

The mental workload is a multidimensional concept [13]. It is defined as the gap between an individual's cognitive resources and mental demands [14, 15], has been widely studied due to its impact on performance across various professions [16–19]. For instance, a study revealed a modest positive association between the mental workload experienced by nurses and their average job performance scores [20]. High mental workload in healthcare has been linked to reduced job performance and increased risks of insomnia, anxiety, and depression among healthcare workers [21]. Similarly, in safety-critical environments like nuclear power plants, excessive cognitive workload has been shown to compromise operator performance and safety [22]. Zahabi et al., [23] pointed out a higher cognitive workload would reduce the speed and accuracy of police officers in performing certain secondary tasks. Multiple studies have also demonstrated a positive association between workload and the occurrence of human error. For instance, Diaz-Piedra et al. [24], conducted research indicating that heightened mental workload is linked to a rise in pilot error. Through an analysis and research of workers in the Mexican manufacturing system, Realyvásquez-Vargas et al., [25] discovered a significant association among the workers' mental workload, physical fatigue, and human errors. The influence of mental workload on personnel performance is evident. Several studies have been conducted to examine the mental workload experienced by individuals in diverse professional environments, with the objective of evaluating their cognitive performance capacities. Törnros and Bolling [26] demonstrated that mobile phone use can occupy a driver's visual, auditory, and cognitive resources, impairing vehicle control and significantly reducing driving performance [27]. When multiple channels are simultaneously engaged, the impact on driving performance is even more pronounced [28]. In fact, even in basic driving conditions, mobile phone usage substantially degrades performance, with the combined occupancy of the visual and cognitive channels resulting in the most pronounced decline in driving ability [29]. These findings underscore the importance of understanding how workload impacts

performance. Studies have shown that prolonged high mental workload significantly increase the likelihood of errors during oceanauts missions [5, 30]. However, there is limited research on multi-channel mental workload in underwater settings. During diving operations, oceanauts need to keep sitting and leaning forward to perform numerous tasks, with minimal movement range in the confined space of a manned submersible. Compared with other work environments, oceanauts face distinct operational challenges. The primary workload for oceanauts is mental workload [31], which remains consistently at a relatively high level [32]. Given these unique operational demands, our study addresses an important gap by examining how visual, auditory, cognitive, and psychomotor workload influences performance in oceanauts. The insights gained may inform training and task design to improve cognitive efficacy in submersible operations.

Previous studies have demonstrated that the cognitive performance of individuals can be impacted by different working conditions. For instance, Sugimoto et al., [33] have examined the impact of single-pedal vehicles on drivers' cognitive workload. Kang et al., [34] compared the influence of standing and continuous sitting in the office on the cognitive performance of workers. Dehais et al., [35] revealed the explanatory mechanism of cognitive failures due to high levels of stress or workload in operational contexts. The impact of cognitive workload on performance in assembly tasks was examined by Biondi et al., [36]. According to Bragg and Redifer's research [37], it was observed that participants exhibited the lowest performance when subjected to the highest level of cognitive load, even when not engaged in a secondary task. Recent studies have also examined how working conditions within submarines affect operators' cognitive performance. Nieuwenhuys et al., [38] investigated the effects of fatigue and sleep deprivation on submariners' cognitive performance. In manned submersible, oceanauts operating in noisy environments with relatively low sound pressure levels (40 dB(A) and 45 dB(A)) exhibited poorer visual performance when they had a lower BMI, compared to those with a normal BMI [1]. Zhu et al. [2], identified extended operating times and sampling challenges as significant factors impacting the cognitive efficiency of oceanauts. However, the patterns of oceanaut's cognitive performance changes in remain unclear. Therefore, it is necessary to investigate the laws governing the cognitive performance changes of oceanauts during manned submersible operations.

The concept of personnel comfort is typically characterised by the absence of suffering and psychological issues, as well as a state of physical relaxation [39–41]. There exists a reciprocal relationship between the personnel comfort and their work efficiency [42]. Numerous

scholarly investigations have been conducted to explore the correlation between employee well-being and job performance. Several studies have examined the negative impact of thermal discomfort induced by high temperature on employee performance [43–45]. In the study conducted by Kosonen and Tan [46], it was observed that the work performance diminishes when the occupants are not in a state of thermal neutrality. The study by Choi et al. [47] indicated that acoustic comfort has a significant impact on job performance. In a study conducted by Cai et al. [48], a significant and robust correlation was observed between subjective measures of perceived comfort and objective measures of cognitive function. Hence, this study examined the impact of human comfort perception on cognitive performance. Considering that manned submersibles are generally operated by a single individual, this study will further investigate the influence of the oceanaut's personal characteristics on cognitive performance.

In recent years, neural networks have been widely employed in cognitive performance research due to their ability to model complex, nonlinear relationships between multiple variables. Studies have demonstrated their efficacy in predicting cognitive workload and task performance in domains such as aviation and automotive safety [49–51]. For example, Hernandez-Sabate et al. [52] employed a convolutional neural network to analyze EEG features linked to pilots' mental workloads during a continuous performance task, which assesses working memory and its performance. Hajinoroozi et al. [53] introduced a Channelized Convolutional Neural Network (CCNN) and its variant to identify instances of poor cognitive performance in drivers. Despite these advances, little research has focused on applying neural networks to predict cognitive performance in underwater settings, where oceanauts operate under unique mental and physical demands.

Given the limited research on multi-channel mental workload among oceanauts, this study aimed to develop a Radial Basis Function model to predict their personalized cognitive performance using four standard cognitive tests. Both the personal characteristics and task attributes were considered to examine the oceanaut's personalized cognitive performance. This paper is organized as follows: “[Method](#)” section presents the development of a mental workload model for typical tasks, which serves as the foundation for the cognitive performance prediction. “[Case study: Construction of Prediction Model for oceanauts' Cognitive performance](#)” section introduces a case study for developing a personalized cognitive performance prediction model for oceanauts, while the results of the study was presented in “[Results](#)” section. “[Discussion](#)” section discusses the implications of the

findings and the corresponding research prospects, and “[Conclusion](#)” section provided the concluding remarks for the paper.

Method

Mental workload model of typical tasks

To understand the impact of mental workload on cognitive performance, we have clarified the mental workload associated with oceanauts' typical tasks. These tasks include: (1) acquiring and evaluating positional information, (2) receiving and executing commands from the interface, (3) processing voice instructions and performing corresponding actions, and (4) responding to inquiries based on received voice information. The construction of the prototypical task process was undertaken. A typical task workload model for personnel has been developed by identifying the resource consumption associated with the four channels of vision, hearing, cognition, and movement.

The VACP (Visual, Auditory, Cognitive, Psychomotor) technique is grounded in Wickens' multi-resource theory [15]. It is a powerful analytical tool for assessing mental workload, initially developed by experts in the aviation field and now applied across numerous domains [54–57]. It predicts the mental workload induced by the task by summing the resource occupied scores across four independent channels: visual, auditory, cognitive, and psychomotor [58, 59]. The estimation of resource occupied in each channel was conducted using VACP Rating Scales (as shown in Table 1). The VACP method comprises three primary steps: (1) Task decomposition. Dividing a complex task into simpler subtasks, with each subtask further segmented into behaviors associated with each of the four channels. (2) Scoring. Applying the VACP scale to evaluate the sub tasks within each channel. By analyzing the task description, the appropriate verbal descriptor is located on the scale, providing a resource consumption score for each channel. (3) Calculating the VACP scores. The VACP score for each subtask is obtained by summing the individual channel scores, and the total VACP score for the complete task is calculated by summing the scores across all subtasks. In the context of a pilot engaging in operational activities, such as inspecting the flight instrument panel, the visual channel resource consumption is quantified as 4 (V3). When the pilot engages in communication with the air traffic controller through verbal exchanges, the auditory channel resource consumption is measured at 4.3 (A4). Furthermore, when the pilot undertakes a complex task comprising multiple sub-tasks, the resource consumption in each channel is determined by adding up the load values associated with each task. During

Table 1 VACP Rating Scales

Visual resource consumption	Visual Description	Auditory resource consumption	Auditory Description
0.0	No visual activity (V0)	0.0	No auditory activity (A0)
1.0	Visually register/detect (detect occurrence of image) (V1)	1.0	Detect/register sound (detect occurrence of sound) (A1)
3.7	Visually discriminate (detect visual differences) (V2)	2.0	Orient to sound (general orientation/attention) (A2)
4.0	Visually inspect/check (discrete inspection/static condition) (V3)	4.2	Orient to sound (selective orientation/attention) (A3)
5.0	Visually locate/align (selective orientation) (V4)	4.3	Verify auditory feedback (detect occurrence of anticipated sound) (A4)
5.4	Visually track/follow (maintain orientation) (V5)	4.9	Interpret semantic content (speech) (A5)
5.9	Visually read (symbol) (V6)	6.6	Discriminate sound characteristics (detect auditory differences) (A6)
7.0	Visually scan/search/monitor (continuous/serial inspection, multiple conditions) (V7)	7.0	Interpret sound patterns (pulse rates, etc.) (A7)
Cognitive resource consumption	Cognitive Description	Psychopsychomotor resource consumption	Psychopsychomotor Description
0.0	No cognitive activity (C0)	0.0	No psychopsychomotor activity (P0)
1.0	Automatic (simple association) (C1)	1.0	Speech (P1)
1.2	Alternative selection (C2)	2.2	Discrete actuation (button, toggle, trigger) (P2)
3.7	Sign/signal recognition (C3)	2.6	Continuous adjustment (flight control, sensor control) (P3)
4.6	Evaluation/Judgment (consider single aspect) (C4)	4.6	Manipulation (P4)
5.3	Encoding/Decoding, recall (C5)	5.8	Discrete adjustment (rotary, vertical thumb wheel, lever position) (P5)
6.8	Evaluation/Judgment (consider several aspects) (C6)	6.5	Symbolic production (writing) (P6)
7.0	Estimation, calculation, conversion (C7)	7.0	Serial discrete manipulation (keyboard entries) (P7)

submarine operations, oceanaut is frequently required to manage complex tasks involving various channels of information. For example, while receiving and executing commands from the interface, the oceanaut must identify interface information (visual channel), memorize manipulation sequence (cognitive channel), and complete continuous operation (psychomotor channel). However, research on multi-channel operational task analyses for oceanaut is limited. Considering that the VACP provides a convenient method for predicting the mental workload associated with specific tasks [60], it was utilized to develop a comprehensive mental workload model for the typical tasks of oceanauts.

This research used VACP scales to construct a comprehensive model of mental workload model for oceanauts' typical tasks. The mental workload associated with four key operational tasks encountered by oceanauts during diving operations was analyzed: (1) acquiring and judging the position information, (2) receiving commands from the interface and executing them, (3) obtaining voice instructions and performing the necessary actions, and (4) receiving voice information and providing responses

to inquiries. The specific calculation process is illustrated in the Fig. 1.

A quantitative model of cognitive performance

The utilisation of speed and accuracy as simplified approaches for performance calculation has been found to be effective [61]. Therefore, the cognitive reaction time and accuracy of the oceanaut were used to construct a specific aspect of the oceanaut cognitive performance model related to problem-solving in this study. The term "reaction time" pertains to the period during which a diver is involved in a particular task. In the context of a diver observing the dive depths of a submersible, the duration required to perceive and process the numerical information can be regarded as an indicator of the diver's reaction time in performing this particular task. Moreover, the precision of the diver's measurement is directly proportional to their level of accuracy.

In order to achieve equitable consideration of both speed and accuracy in the assessment of cognitive performance, the functional model was constructed by drawing upon the Balanced Integration Score (BIS)

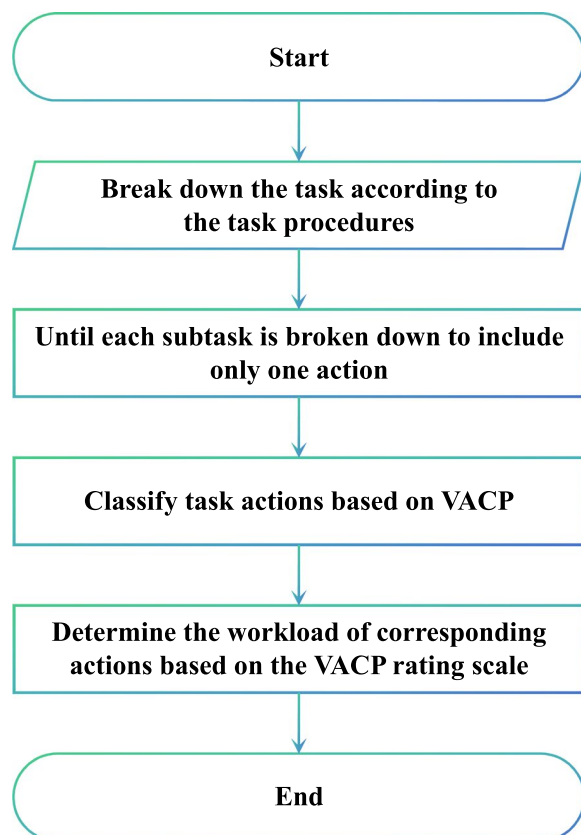


Fig. 1 Calculation process for mental workload based on the VACP scale

model [62] and Speed-Accuracy Score (LISAS) model [63].

A prediction model of cognitive performance with radial basis functions

The Radial Basis Function Neural Network (RBFNN) is a specialized type of feed-forward neural network, characterized by the use of a radial basis function as the activation function in its hidden layer [64]. The RBFNN possesses several advantages, including rapid learning speed, a straightforward architecture, a robust ability to generalise, and a strong capacity for approximation [65, 66]. Some studies have demonstrated the outstanding performance of RBFNN in the prediction of loads and performance [67–69]. Hence, the present study employed a combination of RBFNN and regression analysis to construct a cognitive performance prediction model. We also analyzed the relationships among the cognitive performance of oceanauts under workload in multi-channels, their cognitive performance,

Table 2 Subjective Comfort Scale

Rate	Description
1	Extreme Discomfort
2	Very Discomfort
3	Slight Discomfort
4	No Obvious Comfort or Discomfort
5	Relatively Comfortable
6	Very Comfortable
7	Extremely Comfortable

and subjective comfort. In this way, the study lays a theoretical foundation for improving oceanauts' operational performance and work efficiency.

Case study: construction of prediction model for oceanauts' cognitive performance

Participants

Forty male volunteers in the age range of 21–40 (mean $\pm$ SD = 27.89 $\pm$ 2.13) contributed to the experiment. In terms of educational level, 5 of the participants had an undergraduate degree, 20 held a master's degree, and 15 had a doctorate. The criteria for selection were:

- 1) No chronic pain or discomfort in the neck, upper back, or lower back during the previous 12 months.
- 2) Engaging in daily mental labor in their profession and spending three or more hours per day sitting.
- 3) No history of dyslexia or color blindness.
- 4) Students or professors researching manned submersibles were used to simulate the oceanauts.

In the meantime, all participants were orally informed of the study, and their consent was received.

Apparatus

The participants' cognitive data was recorded by E-prime3.0 software during the experiment, while the Gifcam screen recording software visualized the operation process. The participants were requested to complete the Subjective Comfort Scale, and their subjective comfort levels were evaluated. The Subjective Comfort Scale is a measure of personal comfort with a seven-point Likert scale shown in Table 2.

Tasks

Task 1: Obtaining and evaluating cruising position of the manned submersible. The navigation information interface plays a crucial role for oceanauts oper-

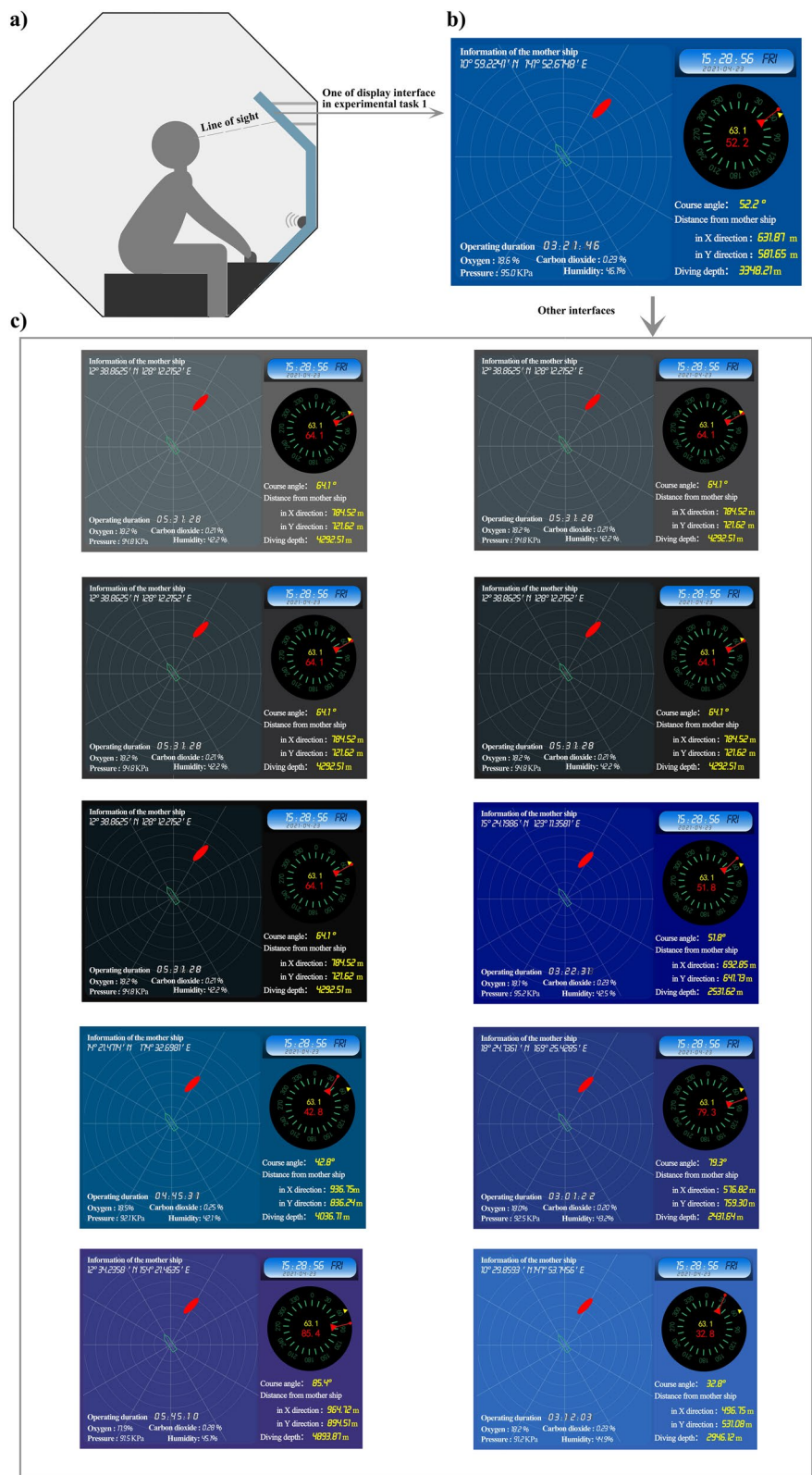


Fig. 2 Experimental task 1 design (a) working posture of the participant (b) the typical display interface in the manned submersible (c) all the interfaces used in the experiment

ating manned submersibles. Participants completed 11 sets of cognitive tests based on a typical display interface (as shown in Fig. 2 (b)) with different color characteristics (as Fig. 2 (c) shown) in a specific posture (as Fig. 2 (a) shown). In each test, the specific test required participants to remember various distance values in the interface. Taking Fig. 2 (b) as an example, it consists of four sub tasks in total: (1) Participants were required to check and memorize the current distance of the manned submersible from the mother ship in the X direction, Y direction distance, heading angle and diving depth data. (2) Afterward, they were instructed to fill up the Subjective Comfort Scale. (3) They need to assess the accuracy of a randomly presented distance value displayed on the interface with information judgment. (4) They were then required to input their judgement outcome using the designated button. Besides, participants' cognitive performance assessed by evaluating their cognitive reaction time and accuracy. To effectively calculate the mental workload, we analyzed the task through task decomposition using the VACP scale.

Task 2: Receiving interface commands and manipulating. The control panel of the manned submersible serves as the principal interface through which the oceanaut acquires alarm information and carries out operational tasks. The upper body posture of the oceanaut during the operating as shown in Fig. 3 (a). This study examines a case involving the management of the operation panel during diving and sampling operations. The operation panel includes control areas for various components such as the "ballast tank," "diving load," "underwater light," "camera," "hydraulic source," and "mechanical arm." Participants were required to complete six different layout features for cognitive tests, as shown in Fig. 3 (c). The task include four sub tasks in total: (1) Participants were required to inspect the interface with operation information and memorize the manipulation sequence depicted in Fig. 3 (b). (2) They need to observe the control panel and perform the actions of hovering and sampling according to the prescribed manipulation sequence. In accordance with the specifications in Fig. 2(b), a total of six buttons with distinct functions need to be pressed. (3) They were required to report their subjective comfort based on their subjective feelings for each panel layout. The duration of time required by the participants to successfully accomplish the assigned task was documented as the fundamental metric for evaluating cognitive performance.

Task 3: Acquisition and Operation of Voice Instruction. The communication system of the manned

submersible plays a vital role in the communication between oceanauts and the mother ship. During the operation, oceanauts frequently encounter situations where they must integrate various forms of communication data in order to successfully accomplish their assigned tasks. Due to the long-term existence of noise and vibration in the manned submersible, participants were required to complete cognitive tests under four different noise frequencies in task 3. It consists of three sub tasks: (1) The participants were required to listen to the voice instructions and memorize the key information. (2) They need to press the appropriate button to complete the corresponding operation. (3) They were asked to provided feedback on their subjective auditory comfort. The cognitive performance of the participants was assessed by measuring their reaction time and accuracy.

Task 4: Obtaining voice information and answering questions. In order for oceanauts to receive instructions from the mother ship during operations, it is crucial to acquire voice information precisely. The participants were required to listen to three distinct sampling frequencies of submersible position data, including heading angle, relative position to the mother ship, and dive depth in task 4. The task include three sub tasks: (1) Participants were asked to listen to the voice instructions and memorize the key information. (2) They need to observe the interface and respond to position-related questions shown in the interface. They should enter the information answer via the keyboard according to the memory. (3) They were required to rate their hearing comfort. Also noted were the participants' reaction times and mental precision.

Analysis for four typical tasks of oceanauts based on VACP is shown in Table 3.

Procedure

During the experiments, the participant was instructed to execute tasks while sitting slightly forward-leaning in the simulated cabin environment. The layout of the display panel, control panel, and communication equipment is shown in Fig. 4. We specifically invited oceanauts with experience in deep-diving missions to evaluate our simulated environment. Their feedback indicated that the simulation scenarios were highly realistic, ensuring the credibility of the study's setup.

Each participant needs to complete a total of 24 tests, which consist of 11 sets of tasks for Task 1, 6 experimental tasks for Task 2, 4 sets of tasks for Task 3, and 3

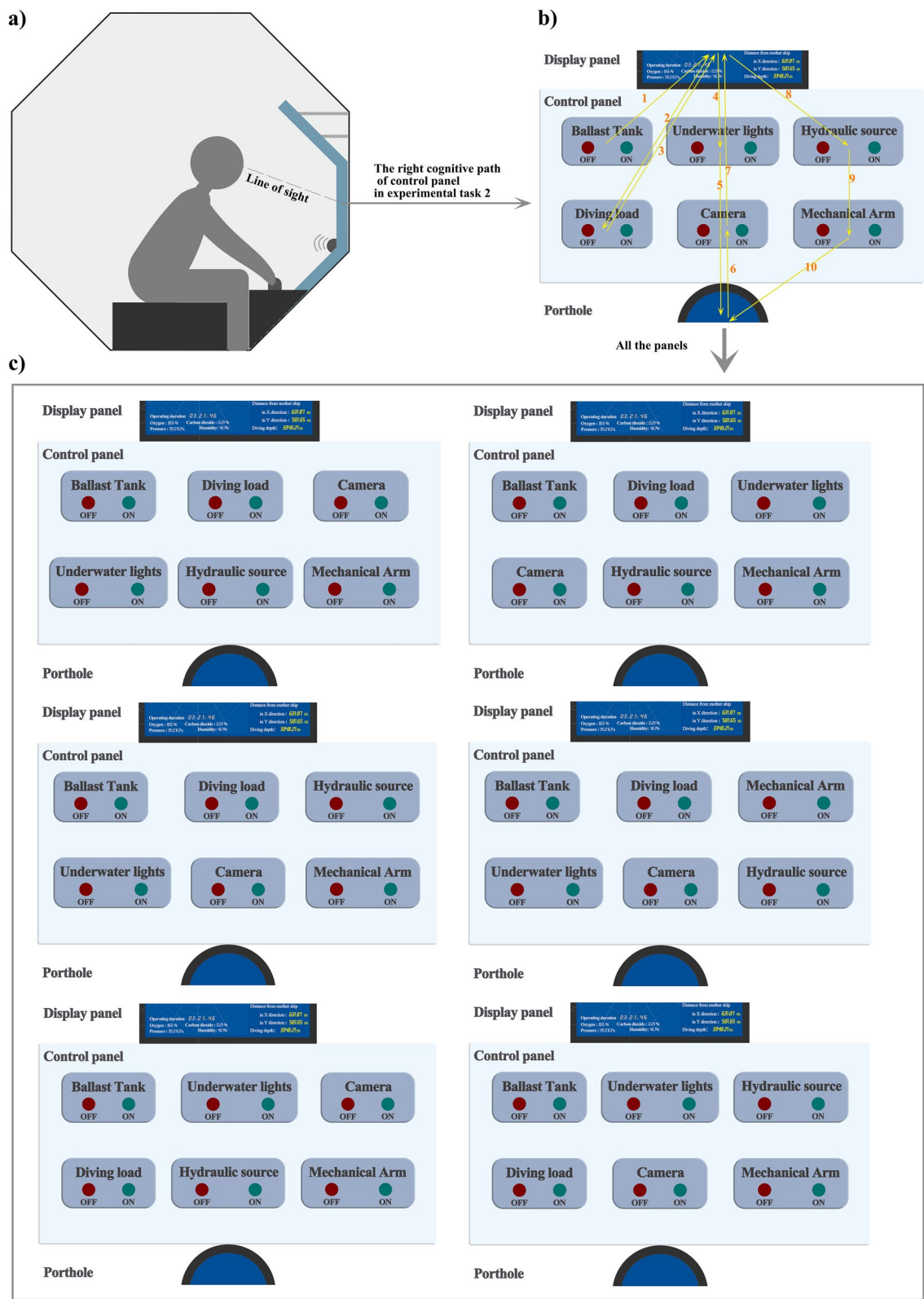


Fig. 3 Experimental task 2 design **(a)** working posture of the participant **(b)** the correct cognitive path **(c)** all the panels used in the experiment

Table 3 Analysis for four typical tasks of oceanauts based on VACP

Task No	Sub Task	Task Type	Activity	VACP Values	Corresponding resource consumption value
1	(1)	Visual	Check navigation information	Visually inspect/check (V3)	4.0
		Cognitive	Memorize navigation information	Encoding/Decoding, recall (C5)	5.3
	(2)	Cognitive	Evaluation of interface visual comfort	Evaluation/Judgment (consider single aspect) (C4)	4.6
	(3)	Visual	Check judgment information	Visually inspect/check (V3)	4.0
		Cognitive	Determine whether the information is correct	Evaluation/Judgment (consider several aspects) (C6)	6.8
	(4)	Psychopsych	Press the appropriate button	Discrete actuation (button, toggle, trigger) (P2)	2.2
2	(1)	Visual	Inspect operation information	Visually inspect/check (V3)	4.0
		Cognitive	Memorize manipulation sequence	Encoding/Decoding, recall (C5)	5.3
	(2)	Visual	Inspect control panel	Visually inspect/check (V3)	4.0
		Cognitive	Match the operation process with the layout design of the control panel	Sign/signal recognition (C3)	3.7
		Psychopsych	Perform the actions (6 times)	Manipulation (P4)	4.6
	(3)	Cognitive	Evaluation of panel visual comfort	Evaluation/Judgment (consider single aspect) (C4)	4.6
3	(1)	Auditory	Listen to the voice instructions	Orient to sound (selective orientation/attention) (A3)	4.2
		Cognitive	Memorize navigation information	Encoding/Decoding, recall (C5)	5.3
	(2)	Psychopsych	Press the appropriate button	Discrete actuation (button, toggle, trigger) (P2)	2.2
	(3)	Cognitive	Evaluation of auditory comfort	Evaluation/Judgment (consider single aspect) (C4)	4.6
4	(1)	Auditory	Listen to the voice instructions	Orient to sound (selective orientation/attention) (A3)	4.2
		Cognitive	Memorize navigation information	Encoding/Decoding, recall (C5)	5.3
	(2)	Visual	Inspect information interface	Visually inspect/check (V3)	4.0
		Cognitive	Think about the answer to question	Evaluation/Judgment (consider several aspects) (C6)	6.8
		Psychopsych	Enter the information answer via the keyboard	Serial discrete manipulation (keyboard entries) (P7)	7.0
	(3)	Cognitive	Evaluation of auditory comfort	Evaluation/Judgment (consider single aspect) (C4)	

experimental tasks for Task 4. In order to mitigate the adverse consequences of the test sequence, we implemented a randomized arrangement of tests for each participant. The participants were informed of the operational methods, the details of the experimental procedure, and the matters needing attention before the test. During the experiment, another participant was seated beside the participant to simulate the natural working environment and operating conditions in the manned submersible. Participants were asked to complete the Subjective Comfort Scale to rate their individual levels of comfort during each test session. Concurrently, they were required to fill out a personal information form, which primarily collected data on age, height, weight,

educational background, and professional experience in years.

Data pre-processing

Multi-channel workload value of oceanauts

The mental workload model for the four typical tasks was analyzed based on the VACP technique as described in “[Mental workload model of typical tasks](#)” section. The analysis results are shown in Table 3. For each task, the resource consumption value for each channel is the sum of the consumption values for that channel across all relevant subtasks. This integration was done to elucidate the overall effort in each channel, as determined by Formula (1):

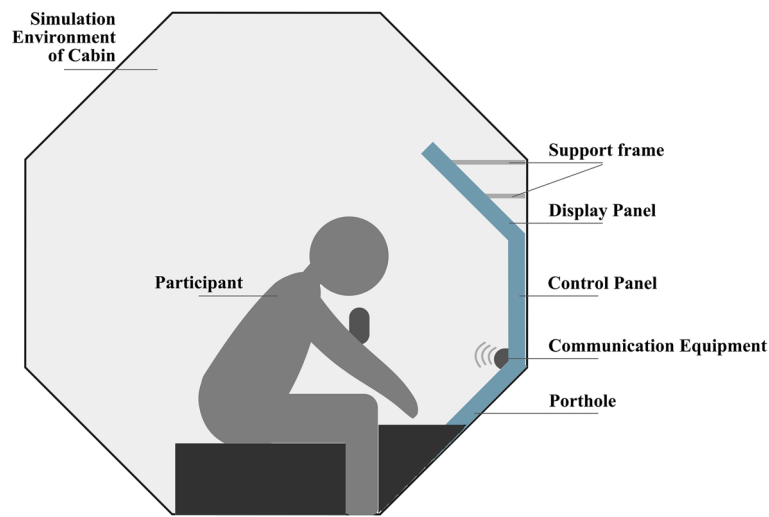


Fig. 4 Schematic diagram of experiment design

$$\begin{cases} L_V = 10 \times \sum_{i=1}^n R_V \\ L_A = 10 \times \sum_{i=1}^n R_A \\ L_C = 10 \times \sum_{i=1}^n R_C \\ L_P = 10 \times \sum_{i=1}^n R_P \end{cases} \quad (1)$$

where R_V is the resource consumption in the visual channel in each sub task, L_V is the workload in vision, R_A is the resource consumption in the auditory channel in each sub task, L_A is the workload in hearing, R_C is the resource consumption in the cognitive channel, L_C is the workload in cognition, R_P is the resource consumption in the psychomotor channel, L_P is the workload in movement.

Taking Task 1 as an example, based on the Table 3 and VACP technique described in “[Mental workload model of typical tasks](#)” section, the resource consumption value for the visual channel is the sum of the VACP values for “visual inspection/check” in sub-tasks (1) and (3) (4+4). According to Formula (1), the workload in the visual channel is 80. Similarly, for the cognitive channel, the resource consumption value is the sum of the VACP values for “Encoding/Decoding,” “Recall (C5),” “Evaluation/Judgment (consider single aspect) (C4),” and “Evaluation/Judgment (consider multiple aspects) (C6)” (5.3+4.6+6.8). According to Formula (1), the cognitive workload is 167. In the motor channel, the resource consumption value corresponds to “Discrete actuation (button, toggle, trigger)” in sub-task (4) with a VACP value of 2.2, resulting in a workload of 22. The calculation method

of workload value for other tasks follows the same procedure, and the final results are shown in Table 4.

Oceanaut’s cognitive performance

Personnel have enhanced cognitive performance when subjected to identical mental accuracy conditions as a result of their reduced reaction time. On the other hand, when subjected to same cognitive reaction time conditions, individuals demonstrate enhanced cognitive performance accompanied with increased cognitive correctness. Reaction time and cognitive accuracy data were gathered from all participants in order to develop a formula, denoted as Formula (2), which represents the cognitive performance of oceanauts for each task. The formula was designed to ensure that performance values are positive and that there is a high degree of discrimination among participants.

$$P_i = \begin{cases} \frac{\overline{CR}}{CR_i} + \frac{\overline{IMG_RT}}{IMG_RT_i} & CR_i \neq 0 \\ \frac{\overline{IMG_RT}}{IMG_RT_i} & CR_i = 0 \end{cases} \quad (2)$$

where IMG_RT_i is the reaction time of i_{th} participant, CR_i is the cognitive accuracy index of i_{th} participant.

Table 4 The workload value of each task using formula 1

Task No	Workload in vision	Workload in hearing	Workload in cognition	Workload in movement
1	80	0	167	22
2	80	0	136	276
3	0	42	99	22
4	40	42	167	70

$\overline{IMG_RT}$ is the average reaction time of all participants, $\overline{CR}$ is the average cognitive accuracy index of all participants (it was recorded as correct = 1 and wrong = 0), while P_i is the initial cognitive performance for i_{th} participant.

To streamline computations and improve the comprehensibility of the model, the outcomes of Eq. 2 are uniformly multiplied by a factor of 10, as demonstrated in formula (3) shown. This ensures that the oceanaut's cognitive performance, along with other input indicators such as age, mental workload in each channel, and subjective comfort, are all represented as integers.

$$CP_i = 10 \times P_i \quad (3)$$

CP_i is the result of the modified cognitive performance for i_{th} participant.

Prediction model of oceanaut's cognitive performance

The new function in the neural network toolbox in the Matlab R2018a software was used to establish a RBFNN model for oceanauts' cognitive performance prediction. The specific process was as follows:

- 1) The RBF networks are constructed based on an output layer, a hidden layer, and an input layer. The Gaussian function was used as the radial basis transformation function (as Formula (4)) to perform non-linear processing on the independent variables, realizing the transformation from the input layer to the hidden layer.

$$h_i = \exp\left(-\frac{1}{2\sigma_i^2}\|X - c_i\|^2\right) \quad (4)$$

where h_i is the output of the i_{th} basic unit in the hidden layer, X is the input vector of the input layer, c_i is the central parameter of the corresponding radial basis transformation function, and σ_i is the width parameter.

- 2) A linear combination of the hidden layer units was performed, and the output layer was obtained. The output function is shown in Formula (5):

$$Y = \sum_{i=1}^m \omega_{ij} h_i \quad (5)$$

where ω_{ij} represents the weight between the unit of i_{th} hidden layer and the unit of j_{th} output.

- 3) The data were randomly divided into a training set and a validation set with no overlap between the two. The unsupervised training method was used to clarify the central parameter and width parameter of

the RBF. the K-means clustering training method was used to train the central parameter c_i of the radial basis function, and the average distance formula was used to solve the width σ_i .

- 4) A supervised training approach was used to solve the weight ω_{ij} , while the basic idea is to seek the smallest distance measure between the actual output Y and the expected output E .

$$\min D = \frac{1}{2} \sum (Y - E) \quad (6)$$

- 5) Network testing and verification: the prediction program was called to carry out data verification.

Due to the participants' academic information obtained in the experiment was the string. It is preprocessed by the assignment method before the model is built:

- The bachelor's degree was assigned a value of 1.
- The master's degree was assigned 2.
- The doctoral degree was set 3.

Results

Outlier detection and data cleaning

The cognitive performance of every participant in each activity was computed using Formula (3). Taking Task 3 as an example, the average reaction time for all participants is 96 s, while Participant 8th reaction time is 38.4 s with the correct answer. Using formulas (2) and (3), the cognitive performance for Participant 8 is calculated as 35. The descriptive statistics of the cognitive performance indicators are shown in Table 5. The presence of outliers in the computation of cognitive performance may be attributed to certain participants who inaccurately manipulated the tasks, leading to shorter reaction times, or who did not diligently undertake the cognitive activities. The inclusion of outliers in the training process of the cognitive performance model can have a detrimental impact on its accuracy. This can result in substantial variances in performance predictions, ultimately diminishing the dependability and usefulness of the predicted outcomes.

Table 5 Descriptive statistics of cognitive performance indicators

Task No	Mean value	Std. deviation	Max	Min
1	36	16.20	64	12
2	24	20.80	56	6
3	30	12.46	42	8
4	22	24.06	48	4

The combination of the Inter-Quartile Range (IQR) approach and the Box-whisker Plot is an effective way to detect the outlier [70, 71]. Therefore, the outlier detection procedure was conducted on the cognitive performance results by employing this method. The IQR is the difference between the upper quartile (Q3) and the lower quartile (Q1), serving primarily as a measure of data dispersion, encompassing 50% of the data. The Q3 corresponds to the 75th percentile, while the lower quartile Q1 represents the 25th percentile. The inner upper fence is calculated as $Q3 + 1.5 \times IQR$, and the inner lower fence as $Q1 - 1.5 \times IQR$. Data points falling outside these fences are classified as outliers. The findings indicated that within the dataset of 960 data points, a total of eight outliers were identified. Subsequently, the prediction model for the cognitive performance of the oceanauts was developed using the remaining 952 data points.

Table 6 Pearson correlation coefficient between comfort evaluation results and various parameters

No	Name of parameter	Correlation coefficient	Sig. (two-tailed)
1	Age	0.186 ^b	0.027
2	Height	0.008	0.948
3	Weight	-0.062	0.550
4	Education Background	0.259 ^b	0.019
5	Professional Experience	-0.113	0.095
6	Visual Workload	-0.467 ^a	0.000
7	Auditory Workload	-0.625 ^a	0.009
8	Cognitive Workload	-0.694 ^a	0.005
9	Motor Workload	-0.524 ^a	0.000
10	Subjective Comfort	0.482 ^b	0.031

^a Correlation is significant at the 0.01 level (2-tailed)

^b Correlation is significant at the 0.05 level (2-tailed)

Training prediction model for oceanauts' cognitive performance

The Pearson correlation coefficient was initially employed to analyze the relationships between the workload of each channel, personal characteristics, and the cognitive performance outcomes of the oceanauts gathered during the experiment. The correlation coefficients between the cognitive performance and these characteristics are shown in Table 6. It can be seen that age, education level, visual workload, auditory workload, cognitive workload, motor workload, and subjective comfort ratings all significantly correlate with cognitive performance outcomes. Thus, the age and educational level of the participant, the workload in vision, hearing, cognition, and movement, and subjective comfort evaluation were used as input. The cognitive performance obtained from Formula (2) was considered as outputs.

During the construction of the prediction model, the calculation of the oceanauts' workload value in each channel was prioritized using Formula (1). Subsequently, all data were input into the RBFNN, divided into a training set and a test set with a ratio of 8:2. The training procedure involved inputting factors such as age, education level, visual workload, auditory workload, cognitive workload, psychomotor workload, and subjective comfort. These inputs were systematically considered, and as a result, the observed cognitive performance of oceanauts was obtained. Table 7 displays several instances of the experimental dataset. Besides, Cross-Validation (CV) is an effective technique for constructing accurate RBFNN models [72, 73]. Therefore, this study used a five-fold cross-validation to assess the model's performance.

For instance, a 35-year-old participant with a master's degree, during Task 1, showed an observed cognitive performance of 35. The RBFNN model predicted the same value of 35, demonstrating perfect prediction accuracy for this individual. Similarly, for a 21-year-old participant with a bachelor's degree, the observed cognitive performance during Task 2 was 9, and the model also predicted

Table 7 Some examples of the experimental dataset

Input characteristics							Output characteristic	Predicted cognitive performance
Age	Educational level	Workload in vision	Workload in hearing	Workload in cognition	Workload in movement	Subjective comfort Results	Observed cognitive performance	
35	2	80	0	167	22	3	32	32
21	1	80	0	136	276	2	10	9
30	3	40	42	167	70	6	36	36
...							...	
25	2	0	42	99	22	7	24	24

9, matching the observed value exactly. These examples illustrate how the model aligns with observed data, and we would further demonstrate the model's overall predictive accuracy by evaluating its performance across the entire dataset.

Validating prediction model for oceanauts' cognitive performance

The Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and Mean Square Error (MSE) are widely used metrics for evaluating the predictive performance of models [74, 75]. Smaller values of RMSE, MAE and MSE indicate lower prediction errors of the model. Thus, the calculation of RMSE, MAE and MSE were performed to evaluate the accuracy of the prediction model.

To demonstrate the efficacy of the predictive model, we compared it to the Back-Propagation Neural Network (BPNN) technique. Based on the same software system and transportation environment, the test sets were predicted. The average values of the cross-validation results are as shown in Table 8. Our results show that the RBFNN model outperforms the BPNN model across all metrics: RMSE=0.219, MAE=0.046, and MSE=0.007 for the RBFNN, compared to RMSE=2.854, MAE=0.628, and MSE=0.362 for the BPNN. These findings clearly indicated that the RBFNN model achieves lower RMSE, MAE, and MSE values, suggesting superior performance in predicting cognitive performance.

Discussion

The predictive model proposed in this study enables effective forecasting of individual cognitive performance in submariners across various operational tasks. This analysis helps clarify how different input characteristics influence the oceanaut's cognitive performance in manned submersibles. To illustrate this, we analyzed the impact of each input variable on the cognitive performance of a 27-year-old male oceanaut with a PhD, using the personalized prediction model we constructed, as shown in Fig. 5.

Except for the visual channel workload, all other variables were held constant, the variations in the oceanaut's cognitive performance were taken into account. Fig. 5 (a) shows that within the range of 0–80, the oceanauts'

cognitive performance improves as the visual channel workload increases. However, when the workload value on the visual channel exceeds 80, the cognitive performance decreases with the visual workload. In order to optimize cognitive performance, it is recommended that the workload placed on the visual channel of oceanauts falls between the range of 80–90.

When examining the impact of auditory channel burden on cognitive performance, it was observed that the cognitive performance of oceanauts exhibits an increase within the range of 0–50 auditory workload, as depicted in Fig. 5 (b). When the auditory workload exceeds 50, the cognitive performance declines as the workload increases. Therefore, maintaining the auditory workload within the range of 40–50 can effectively sustain cognitive performance at its optimal level.

When comparing the burden of the visual and auditory channels, it can be observed that the cognitive performance of oceanauts improves when the range of cognitive workload channels broadens, as depicted in Fig. 5 (c). Nevertheless, it has been observed that when the cognitive stress surpasses a threshold of 120, the cognitive performance of oceanauts starts to exhibit a deterioration. Therefore, it is recommended to maintain the cognitive workload within the range of 120–130.

When the workload of the psychomotor channel approaches 30, there is a corresponding drop in cognitive performance. The data presented in Fig. 5 (d) demonstrates a clear correlation between the progressive rise in psychomotor effort and the subsequent decline in cognitive function.

These findings have also been corroborated in prior research. Multiple research have provided evidence that inadequate and excessive workload can have detrimental effects on staff performance, resulting in neglect, errors, and reduced cognitive function [16, 76]. Paxion et al. [77], highlighted that there is a correlation between the rise in workload and the decline in staff, which subsequently affects driving performance. The study conducted by Melnicuk [78] utilized autonomous persons driving as the research context and discovered a negative impact of non-optimal workload levels on drivers' performance. Hence, it is not feasible to immediately decrease the workload value as it may not necessarily enhance the cognitive performance of individuals.

Concerning comfort, we discovered that the cognitive performance of oceanauts increases within a specific comfort range. However, cognitive performance would decline under extremely comfortable conditions. Using the seven-level comfort scale employed in this study as an illustration, cognitive performance was highest when the participants' perception of comfort reached levels 5 (relatively comfortable) and 6 (very comfortable).

Table 8 The average values of the cross-validation results of RBFNN and BPNN method

Model	Prediction error		
	RMSE	MAE	MSE
RBFNN	0.219	0.046	0.007
BPNN	2.854	0.628	0.362

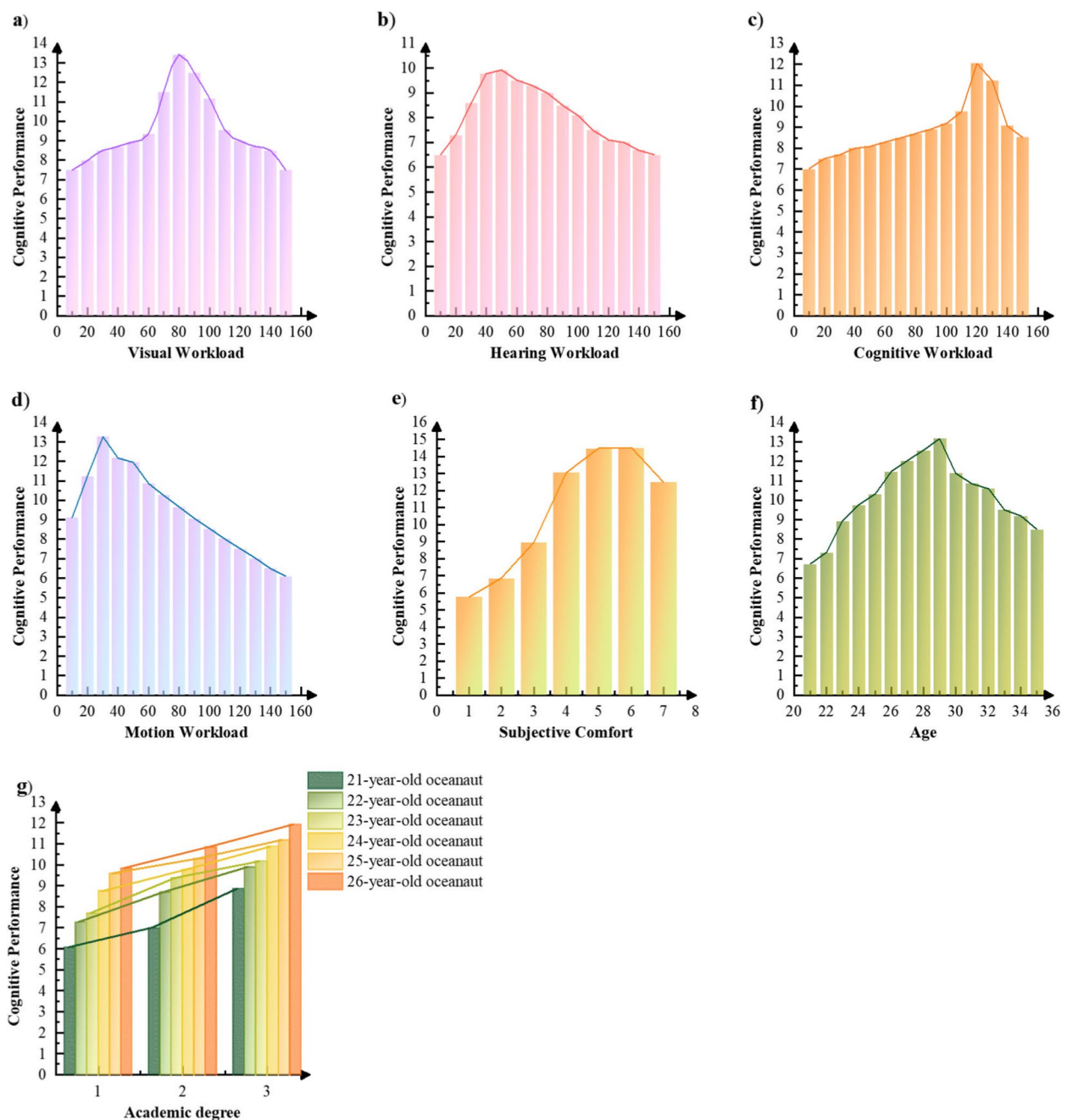


Fig. 5 The influence of various variables on cognitive performance (a) Visual workload (b) Hearing Workload (c) Cognitive Workload (d) Motion Workload (e) Subjective Comfort (f) Age (g) Academic degree

However, when the participant was exceedingly satisfied, neither comfort nor discomfort was evident [79] studied the thermal environment. They discovered that a decrease in thermal comfort would have a negative effect on the cognitive performance of personnel, which is consistent with this study's conclusion. Other investigations have produced comparable results. According

to the study conducted by Tian et al., [80], it was determined that individuals see the personnel in a setting with a temperature of 26 °C as pleasant and acceptable. However, when the temperature rises to 39 °C, the personnel is perceived as slightly uncomfortable and unsatisfactory. Simultaneously, it has been shown that the cognitive abilities of individuals operating in an environment

with a temperature of 39°C exhibit a notable decline compared to those operating in a 26 °C environment. Furthermore, individuals functioning in a relatively pleasant atmosphere characterized as "slightly warm" (28 °C) have inferior cognitive performance than those in a comfortable setting at 24 °C [81]. These results demonstrate the dependability of this study's findings.

As depicted in Fig. 5 (f), the constructed prediction model used the same workload values to examine the effect of age. Based on age analysis, the oceanaut with the highest cognitive performance was 29 years old, while the participant with the lowest cognitive performance was 21 years old. These outcomes can be explained by the increased demands placed on the physical functions and reactivity of personnel. In China, the designated age range for oceanauts was 22 to 35 years old.

Additionally, we took into account the influence of individuals' educational background on their cognitive ability, as Fig. 5 (g) shown. It is evident that there exists a positive correlation between the academic level attained and the cognitive performance of the oceanaut. When combined with the findings of Kremen et al., [82] it is shown that education before the age of 20 might greatly contribute to the formation of cognitive reserve and cognitive testing performance later in life. At the same time, cognitive reserve research has found that a higher education level is connected with an increase in cognitive reserve [83], and studies have demonstrated that the development of cognitive reserve can lead to higher cognitive capacity [84]. These data are consistent with the study's conclusions. Besides, it also can be seen that even with the same education level, 21-year-old oceanauts demonstrate relatively lower cognitive efficiency, consistent with the predictions shown in Fig. 5 (f).

Based on the aforementioned comprehensive analysis, the cognitive performance of oceanauts can be enhanced by defining a reasonable value range for each variable. Specifically, the visual channel workload is deemed optimal within the range of 80–90, the auditory channel workload within 40–50, the cognitive workload channel within 120–130, and the psychomotor workload channel within 30–40. The enhancement of cognitive performance can be attributed to the presence of relatively comfortable conditions pertaining to vision and hearing, which contribute to overall comfort. Nevertheless, too comfortable settings are not conducive to the cognitive performance of oceanauts. It also has been discovered that the oceanauts' cognitive performance is positively influenced by factors such as an appropriate age range and a better degree of education. In addition to task-specific settings and personnel selection strategies aimed at enhancing oceanauts' cognitive performance, future research could examine a wider range of

cognitive performance improvement techniques, such as methods of physiological rhythm regulation, attentional deployment conditioning, and response modulation. For instance, stress has been shown to impact cognitive performance in driving contexts. Béquet et al. [85] analyzed the effectiveness of haptic, breath-guided techniques for stress regulation in driving, emphasizing the importance of accounting for individual and contextual differences in such regulation strategies. The wrist-worn mobile heart rate regulator, *ambienBeat*, has been studied for its potential to indirectly improve cognitive performance [86]. A comprehensive literature review has explored a variety of strategies (encompassing context selection, attentional deployment, and response modulation et al.) to facilitate mindless stress management within Advanced Driver Assistance Systems (ADAS) [87].

Additionally, several studies have demonstrated that physiological indicators and personality traits significantly impact cognitive efficacy [88–91]. Evin et al. [91] utilized physiological and behavioral data combined with machine learning techniques to classify drivers' personality traits across various driving environments, laying a foundation for studying driver capability and condition. Zhu et al. [8] analyzes the relationship between the physiological parameter and operator's visual comfort in the enclosed cabin. Examining the influence of these factors on oceanaut's cognitive efficacy would be a key focus in future research. Additionally, this study has some limitations, such as the participants not being actual operators in the manned submersible. In future studies, we plan to invite actual oceanauts of manned submersible to conduct simulation experiments or verify our conclusions in the real-world operating environment of a manned submersible.

Conclusion

The collection of data pertaining to the cognitive performance of participants was accomplished through the implementation of a simulated mission experiment. Several variables were examined, encompassing multi-channel workload, educational background, age, and subjective comfort. The primary findings can be summarized as follows:

- (1) This paper proposes a personalized cognitive performance prediction model for oceanauts, utilizing Radial Basis Functions as the underlying framework. The model's predicted data yielded an RMSE of 0.219, MAE of 0.046, and MSE of 0.007, which are lower than those of other models. These results demonstrate the model's effectiveness in correlating personnel's multi-channel duties with their cognitive performance.

- (2) Oceanaut's cognitive performance can be enhanced by a reasonable burden for each channel. This study determined the optimal channel burden values, which varied between channels. It is recommended to maintain the visual workload at 80–90, the auditory workload at 40–50, the psychomotor workload at 30–40, and the overall cognitive load at 120–130 to optimize cognitive performance.
- (3) Subjective comfort, age appropriateness, and academic level all influence the oceanauts' cognitive performance. This study found that cognitive performance peaked when participants reported feeling relatively comfortable or very comfortable. The highest cognitive performance was observed in a 29-year-old oceanaut, and those with PhDs demonstrated relatively higher cognitive performance overall. Therefore, optimizing design for comfort, selecting candidates within an appropriate age range, and ensuring a sufficient level of education are critical considerations for the selection and task organization of oceanauts. This study provides a preliminary foundation for improving oceanaut's cognitive performance.

The results of this study are of significant relevance to the task organization and selection process of oceanauts engaged in submarine operations. In the subsequent phase of our investigation, we intend to generate a matrix comprising workload values across several channels that are optimized for enhancing cognitive performance. This endeavor aims to establish a comprehensive theoretical framework for the advanced functioning of the oceanaut.

Abbreviations

CCNN	Channelized Convolutional Neural Network
VACP	Visual, Auditory, Cognitive, Psychomotor
BIS	Balanced Integration Score
LISAS	Speed-Accuracy Score
RBFNN	Radial Basis Function Neural Network
IQR	Inter-Quartile Range
CV	Cross-validation
RMSE	Root Mean Square Error
MAE	Mean Absolute Error
MSE	Mean Square Error
BPNN	Back-Propagation Neural Network
ADAS	Advanced Driver Assistance Systems

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Authors' contributions

MZ conceived of the study, designed the experiment and initiated the research. JW and XZ carried out the execution of experiments and the gathering of data. DC is the funding recipient. All authors read and approved the final manuscript.

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Data availability

Data will be made available upon written request to the corresponding author and with the execution of a data use agreement.

Declarations

Ethics approval and consent to participate

All participants completed written informed consent. The manuscript is approved by medical ethics committee of Ningbo University. All procedures are conducted in accordance with the Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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